



ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH
TRƯỜNG ĐẠI HỌC KINH TẾ - LUẬT

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
UNIVERSITY OF ECONOMICS AND LAW

DATA MINING

Lecture 1: Introduction to Data Mining

EMA4002 | Compulsory | 03 credit



Lecture 1 outline

Section	What we cover
1.1.	Why data mining now, what it is, and where it sits in the KDD process
1.2.	The kinds of data that can be mined, from relational tables to text and networks
1.3.	Data mining functionalities: the six families of pattern
1.4.	How data mining systems are classified, and the disciplines behind them
1.5.	Major issues: methodology, interaction, scalability, privacy
Lab.	Guided lab in Google Colab

Lecture 1 roadmap and outcome

1

Explain what data mining is, and place it correctly inside the knowledge discovery process that runs from raw data to a decision

CLO1

2

Describe the kinds of data an economics analyst meets at work, and say what each kind allows you to ask

CLO1

3

Distinguish the main mining functionalities, and match a business question to the functionality that answers it

CLO2

4

Open a real dataset in Google Colab and produce a first honest description of what is inside it

CLO3

From the connection with Statistics, Econometrics and Programming

What we did in the opening example	Where it sits in Chapter 1
Asked what a fair house price is, using real listings	What data mining is (1.1)
Described the data, ran correlations, spotted odd rows	Pre-processing inside the KDD process (1.1)
Explained with OLS coefficients, predicted with a forest	Descriptive and predictive mining (1.3)
Combined statistics, econometrics and programming	Confluence of disciplines (1.4)
Met overfitting: the simple tools ran out of road	Why data mining, why the confluence (1.1, 1.4)
Regression, trees and clustering still ahead of us	Mining functionalities (1.3)

You have already done data mining by hand. This lecture is the map of what you did, and of everything that comes next.

1.1

Introduction to Data Mining

Why now, what it is, and the process it belongs to

1.1 Introduction to Data Mining

We are drowning in data and starving for knowledge

The data explosion

- Storage moved from megabytes to terabytes, and then to petabytes, inside a single working career
- Collection is now automatic: point-of-sale terminals, bank cores, tax portals, sensors, web logs
- Every transaction that once left a paper trail now leaves a row in a table
- Reading it is no longer an option, so the analysis has to be automated

Where the abundance comes from

- Business: e-commerce baskets, card transactions, stock ticks, supply-chain events
- Science: remote sensing, genomics, the output of large simulations
- Society: news, social platforms, mobile phones, cameras on every street
- Government: tax portals, customs records, statistical registers, open data

THE ARGUMENT IN ONE LINE

Necessity is the mother of invention. Data mining is the automated analysis of data sets far too large for anyone to read, and the discipline exists because the data arrived before the methods did.

1.1 Introduction to Data Mining

Four eras of producing knowledge

1. Empirical science

Before 1600

Describe what you observe and record it carefully. Knowledge is a catalogue of observations.

2. Theoretical science

1600 to 1950s

Every discipline grows a theoretical branch. Models explain the observations and predict new ones.

3. Computational science

1950s to 1990s

Simulation takes over where closed-form solutions run out. A third branch appears in physics, ecology, linguistics.

4. Data science

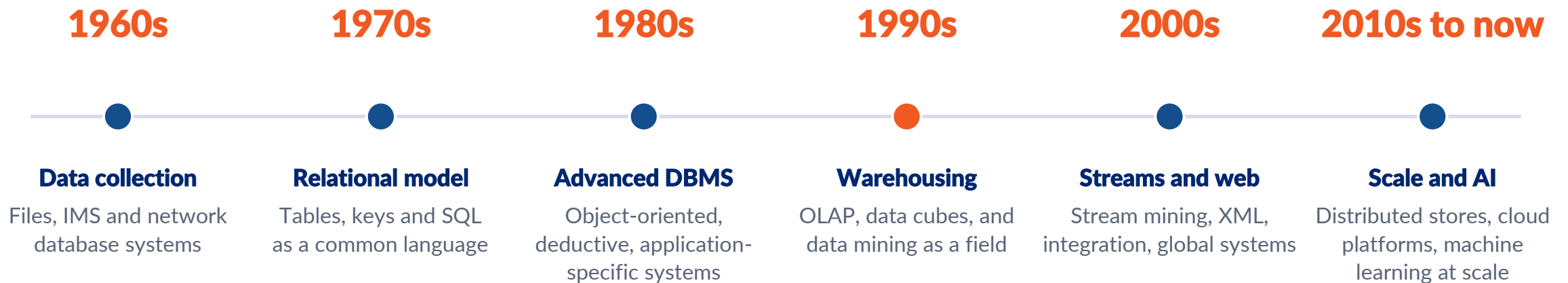
1990s to now

Instruments and information systems generate data faster than theory can consume it. Discovery starts from the data.

After Jim Gray and Alex Szalay, Comm. ACM 45(11), 2002. The fourth era is where this course lives.

1.1 Introduction to Data Mining

Data mining is the child of database technology



Data mining became possible only once the data was already stored, integrated and query-ready. The warehouse came first the mining came after.

1.1 Introduction to Data Mining

What is data mining?

DEFINITION

Data mining is the extraction of interesting patterns or knowledge from a large amount of data. A pattern qualifies as interesting when it is non-trivial, implicit, previously unknown and potentially useful.

Non-trivial

It could not be obtained by a single lookup

Implicit

It is present in the data without being recorded

Previously unknown

Nobody in the firm could state it beforehand

Potentially useful

Somebody could act on it

A misnomer and its aliases

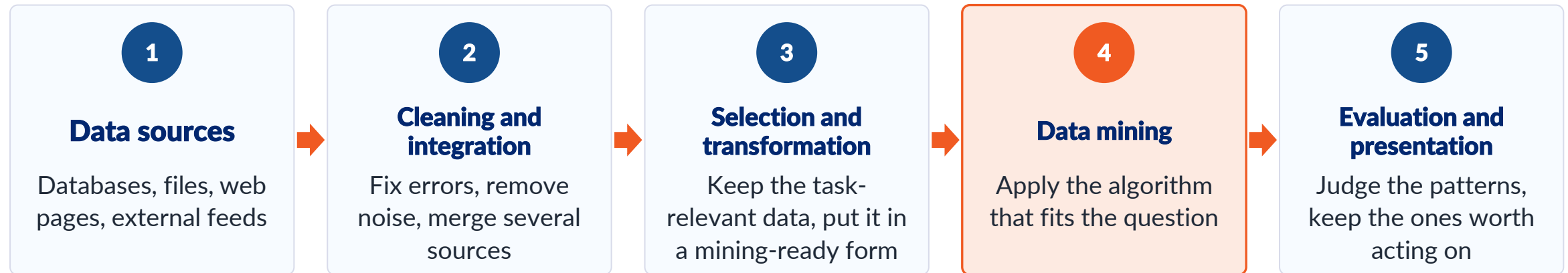
- We mine knowledge out of data, so gold mining would be named after the gold. The name stuck anyway
- Also called knowledge discovery in databases (KDD), knowledge extraction, pattern analysis, information harvesting, business intelligence

Is everything data mining?

- A search engine query is retrieval, not discovery
- A SQL report answers a question you already knew how to ask
- A deductive expert system applies rules a human wrote

1.1 Introduction to Data Mining

The knowledge discovery (KDD) process



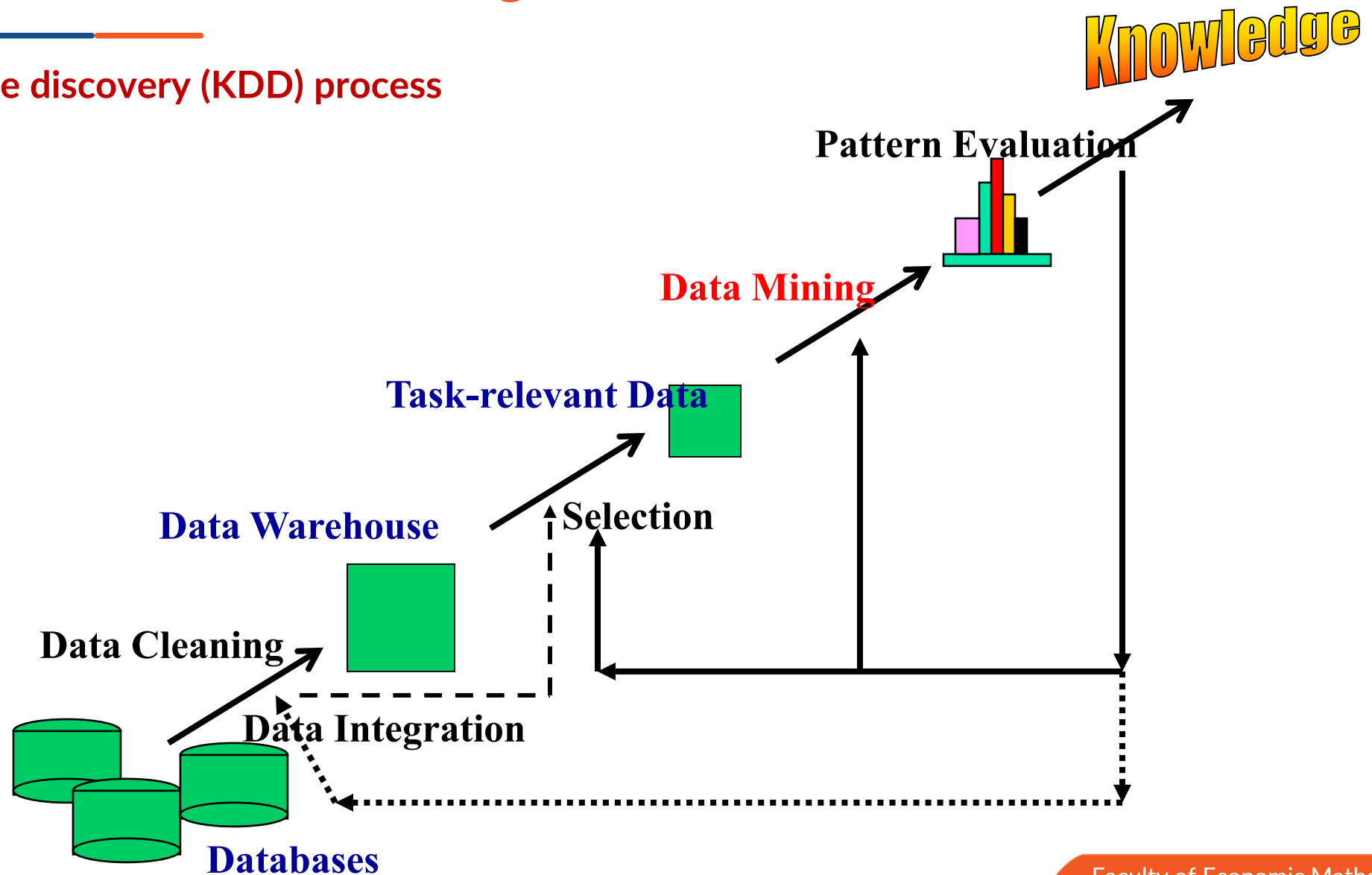
The output is knowledge: a statement about the business that somebody can defend, and act on

WHERE THE TIME ACTUALLY GOES

Steps 2 and 3 consume most of a real project. Practitioners routinely report that preparing the data takes longer than every modelling step combined. Chapter 2 is devoted to that work, so treat it as the core of the course rather than as a preliminary.

1.1 Introduction to Data Mining

The knowledge discovery (KDD) process



1.1 Introduction to Data Mining

One process, two vocabularies

Database and warehouse community

- Databases and other sources
- Data cleaning and data integration
- Data warehouse
- Selection of task-relevant data
- Data mining
- Pattern evaluation, then knowledge

Machine learning and statistics community

- Input data
- Pre-processing: integration, normalisation, feature selection, dimension reduction
- Data mining: association, classification, clustering, outlier analysis
- Post-processing: evaluation, selection, interpretation, visualisation
- Information, then knowledge

The two lists describe the same pipeline. One is written by people who own the data, the other by people who own the algorithms. Read a paper and you can usually tell which side wrote it.

Medical data mining is a familiar example of the second view: pre-process and reduce dimensions, classify or cluster, then post-process for presentation.

1.1 Introduction to Data Mining

Data mining inside business intelligence



1.1 Introduction to Data Mining

Query, report, or mine? A House dataset decides

Query

“How many houses in Thủ Đức sold below 5 tỷ last quarter?”

- One SQL statement over one table
- You know the answer's shape before you run it
- The system retrieves, it does not discover

OLAP report

“Median price per m² by district by quarter, drilled from district to ward.”

- Aggregation over a data cube
- You choose the dimensions and the level
- Powerful, and still driven by your hypothesis

Data mining

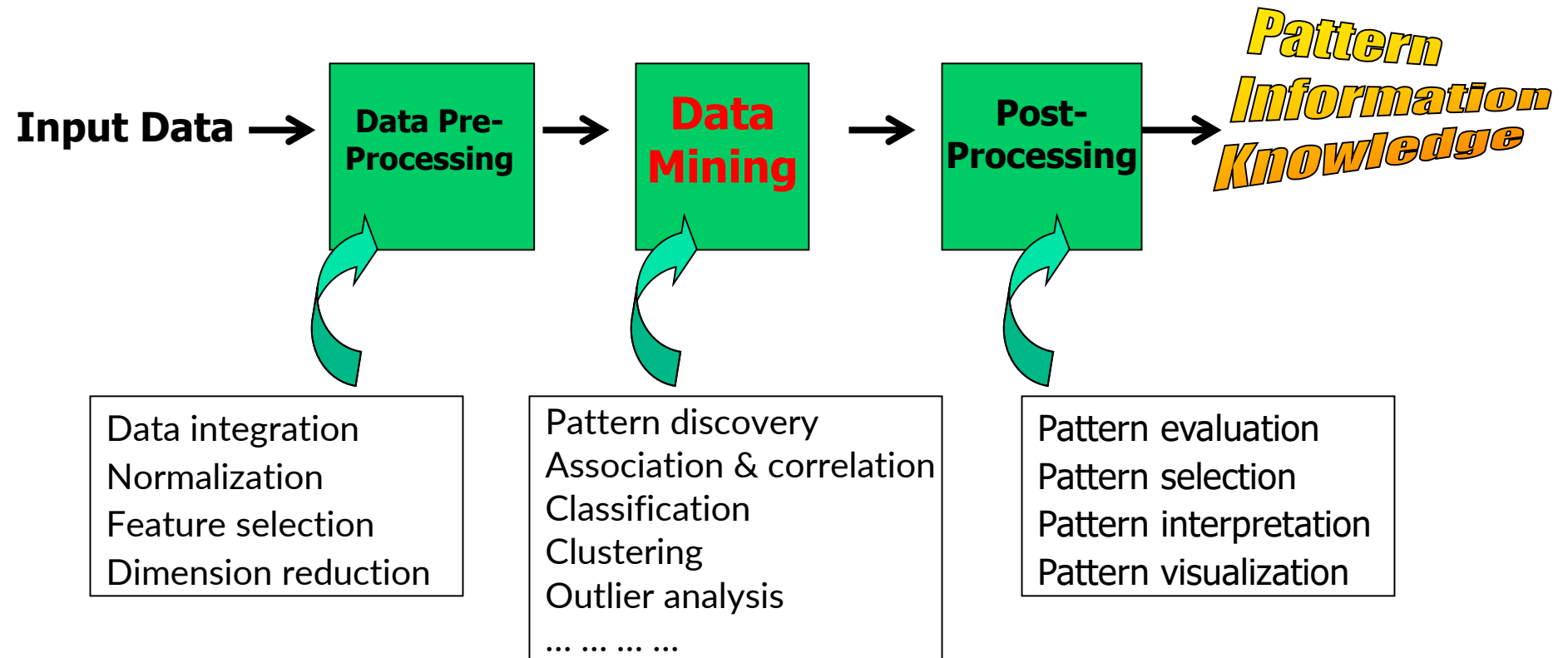
“Which combinations of area, distance to centre and legal status explain price, and which listings are priced abnormally?”

- The pattern is not specified in advance
- The algorithm searches a space you cannot enumerate
- The result may surprise the manager

The dividing line is who supplies the hypothesis. When the analyst supplies it, the machine verifies. When the data supplies it, the machine discovers.

1.1 Introduction to Data Mining

KDD Process: A Typical View from ML and Statistics



Working example: the Saigon Bean chain

The business

- **Saigon Bean** is a fictional coffee chain with **12 stores** across Ho Chi Minh City: flagship, mall, campus, airport, drive-thru
- 17 products in 4 categories: coffee, tea, pastry, merchandise
- Three sales channels: in-store, mobile app (loyalty ID), delivery
- Head office wants to grow from 12 to 30 stores by 2028

The data (June to August 2026)

- **saigon_bean_transactions.csv**: 97,727 order lines, 51,253 orders
- **saigon_bean_stores.csv**: 12 stores, district, seats, store type
- **saigon_bean_products.csv**: 17 products with category and list price
- Total revenue about 5.6 billion VND in three months

Questions the head office is asking

- Which products are bought together, and should the menu be bundled?
- Which customers are about to stop visiting?
- How should stores be grouped for marketing?
- Which store had an unusual August, and why?

Why data mining? We are drowning in data

12

stores

557

orders per day

51,253

orders in 3 months

97,727

receipt lines

Where the data comes from

- Point-of-sale receipts, every line of every order
- Mobile app: loyalty ID, time, location, discount
- Delivery platforms, customer reviews, social media
- Suppliers, staff rosters, weather, local events

Why Excel and SQL are not enough

- Automated collection produces far more than anyone can read
- A query answers a question you already know how to ask
- The valuable patterns are implicit and previously unknown
- Necessity is the mother of invention: automated analysis of massive data

We are drowning in data, but starving for knowledge (Han, Kamber & Pei)

Practice 1

- 1 Label each task as a query, a report or a mining problem, and justify each answer in one line. (a) Total revenue of store Q1 last week. (b) The five products most often bought together with an iced latte. (c) Revenue by district by quarter. (d) The stores whose daily pattern resembles no other store.
- 2 Take one question from your own field of interest, in finance, marketing, public policy or logistics. Write it as a KDD pipeline: which data you would need, what cleaning it would demand, which mining task answers it, and how you would decide whether the answer is any good.
- 3 Two analysts run the same algorithm on the same table and report different findings. Give two reasons from section 1.1 that could explain the difference, without either of them making an arithmetic mistake.

1.2

Data Mining on Different Data Types

What can be mined, and what each kind of data lets you ask

1.2 Data Mining on Different Data Types

The three sources you will actually meet at work

Relational database

- Tables linked by keys, one row per entity
- The operational backbone of any firm
- Saigon Bean: a customer table, a product table, an order table
- Mining here needs joins before it needs algorithms

Data warehouse

- Integrated, subject-oriented, historical, non-volatile
- Organised by dimension: time, store, product, region
- Saigon Bean: a sales cube already cleaned and conformed
- Chapter 2 of this course builds one

Transactional database

- One record per transaction, each holding a set of items
- No fixed number of columns per row
- Saigon Bean: one row per receipt, listing what was on it
- The natural input to association rule mining

These three cover most of what an economics graduate is handed on the first day. The remaining types matter when the question involves time, place, language or relationships.

1.2 Data Mining on Different Data Types

Six advanced kinds, and the questions they open

1 Streams and sensor data

Ordered, time-varying, potentially infinite. Card authorisations, tick data, traffic counters. You see each record once.

2 Time series and sequences

CPI, exchange rates, a customer's ordered purchases, a patient's visits. Order carries the meaning.

3 Graphs and networks

Interbank exposure, supply chains, ownership structures, social ties. The edges are the data.

4 Spatial and spatiotemporal

Store locations, land prices, ride-hailing traces. Distance and neighbourhood change what similar means.

5 Text and the web

Annual reports, news, product reviews, search logs, clickstreams. Meaning has to be extracted before it can be mined.

6 Multimedia and legacy

Product images, store camera video, scanned archives, old systems nobody dares replace.

The data type decides which questions are answerable and which algorithms are admissible. Choose the algorithm after you have looked at the data, never before.

1.3

Data Mining Functionalities

The six families of pattern, and the questions each one answers

1.3 Data Mining Functionalities

Two purposes, one dataset

Descriptive mining

- Characterise what is already in the data
- The output is a summary a person can read and argue with
- Tasks: characterisation, discrimination, association, clustering
- Judged by whether it is compact, accurate and informative

Predictive mining

- Infer a value or a label that the data does not contain
- The output is a model, judged on cases it has never seen
- Tasks: classification, numeric prediction, outlier scoring
- Judged by measured performance on held-out data

THE SAME TABLE SERVES BOTH

Saigon Bean's receipts describe which items travel together, which is descriptive. The same receipts predict which customer will stop coming next month, which is predictive. The question decides the task, and the data rarely decides anything.

Han, Kamber & Pei call this the knowledge dimension of the multi-dimensional view. It returns in section 1.4.

1.3 Data Mining Functionalities

(1) Characterisation and discrimination

Characterisation

- Summarise a target class in general terms
- Built by generalisation: roll individual records up to a concept level
- Attribute-oriented induction and OLAP roll-up do the work

CHARACTERISATION AT SAIGON BEAN

Loyalty members buy on average 2.4 times a week, spend 58,000 dong per visit, and 71% of their visits fall between 07:00 and 09:30.

Discrimination

- Contrast a target class against one or more comparison classes
- The output is a difference, not a portrait
- Answers the question a manager actually asks: how do these two groups differ

DISCRIMINATION AT SAIGON BEAN

Customers who stopped coming in Q2 differ from those who stayed mainly in one respect: they almost never ordered food alongside a drink.

1.3 Data Mining Functionalities

(2) Frequent patterns, association and correlation

What is being searched for

- A frequent itemset is a set of items that appear together often
- A rule has the form $X \rightarrow Y$, read as: receipts holding X tend to hold Y
- Support and confidence quantify how far the rule can be trusted

READ THE TWO NUMBERS SEPARATELY

Support says the rule covers enough of the business to be worth a manager's attention. Confidence says the rule holds when the antecedent occurs. A rule can be strong on one and useless on the other.

```
iced_latte → croissant  
           [support 3%, confidence 62%]
```

```
support      = share of ALL receipts holding both  
confidence   = share of latte receipts that  
               also hold a croissant
```

```
diaper → beer [0.5%, 75%]
```

THREE WORDS THAT ARE NOT SYNONYMS

Association is co-occurrence. Correlation is a statistical relationship. Causality is a claim about what would happen under intervention. Lift and the chi-square test separate the frequent from the interesting, and neither of them delivers causality.

1.3 Data Mining Functionalities

(3) Classification and numeric prediction

How it works

- Supervised learning: the training records already carry the label
- Build a model from labelled examples, then apply it to new records
- When the target is a number instead of a label, the task is numeric prediction
- Accuracy is measured on data the model has never seen

Typical methods

- Decision trees and rule-based classifiers
- Naive Bayes and Bayesian networks
- Logistic regression, k-nearest neighbours
- Support vector machines, neural networks, ensembles

Credit scoring

Will this applicant repay?

Churn prediction

Which customer leaves next month?

Fraud detection

Is this transaction genuine?

Audit selection

Which return deserves inspection?

Econometrics gave you regression with a story about the coefficients. Classification asks a narrower question and answers it with a measured error rate.

1.3 Data Mining Functionalities

(4) Cluster analysis

Unsupervised by definition

- No label exists. The groups are what you are looking for
- Principle: maximise similarity inside a cluster, minimise it between clusters
- Everything therefore depends on how similarity is defined
- The analyst still has to name the clusters and defend the names

Where economists use it

- Customer segmentation for pricing and promotion
- Store or branch typology
- Grouping provinces or markets before modelling each group
- Portfolio grouping, and grouping firms before a panel regression

SAIGON BEAN, TWELVE STORES

Clustering on hourly sales profiles splits the twelve stores into three groups: an office-district type peaking before 09:00, a campus type with a long flat afternoon, and a residential type that only comes alive at the weekend. Nobody labelled the stores in advance. The grouping came out of the data, and the three names came out of the analyst.

1.3 Data Mining Functionalities

(5) Outlier analysis

DEFINITION

An outlier is a data object that does not comply with the general behaviour of the rest of the data.

Noise or treasure?

- In quality control you remove it
- In fraud detection it is the entire point
- One person's garbage is another person's treasure

How they are found

- As a by-product of clustering or regression
- Distance-based and density-based methods
- Statistical models of what normal looks like

Where it earns its keep

- Card fraud and money laundering
- Meter tampering and tax audit selection
- Market manipulation and rare-event analysis

Before deleting a strange row, find out why it is strange. In the house-price example the most interesting listings were the ones the model got wrong.

1.3 Data Mining Functionalities

(6) Order and structure: sequences, trends, graphs, the web

Time and order

- Trend, time series and deviation analysis, including regression and value prediction
- Sequential patterns: buy a phone first, then a case, then a screen protector
- Periodicity analysis, and similarity search between sequences
- Stream mining, where the data is ordered, time-varying and potentially infinite

Structure and links

- Graph mining: frequent subgraphs, trees, recurring substructures
- Information networks: actors as nodes, relationships as edges
- Interbank exposure, ownership chains, co-authorship, supply networks
- Web mining: from PageRank to opinion mining and usage mining

Order and structure carry information that a flat table throws away. Flatten a sequence into totals and you have discarded the part a manager was going to act on.

These families sit in the later chapters of the textbook. They appear here so that the map of the field is complete.

1.3 Data Mining Functionalities

Are all mined patterns interesting?

Objective measures

- Support, confidence and lift for rules
- Accuracy, precision and recall for classifiers
- Coverage: how much of the data the pattern speaks about
- Statistical significance, which large data makes easy to obtain and easy to misread

Subjective measures

- Novelty: did it surprise anyone who knows the business
- Actionability: can the firm do something about it
- Timeliness: does it arrive while the decision is still open
- Typicality against novelty, which pull in opposite directions

THE HARDER GOAL

Evaluating patterns after mining is the easy version. The research goal is to push the interestingness constraint into the search itself, so that only interesting patterns are generated. Constraint-based mining returns in Chapter 3.

A pattern that nobody acts on is a cost, not a discovery.

Practice 2

- 1 Match each business question to one functionality among characterisation, discrimination, association, classification, clustering, outlier analysis and sequence analysis. (a) Which provinces behave alike in their consumption pattern? (b) Which applicants will default? (c) What distinguishes customers who left from those who stayed? (d) Which supplier invoices deserve a second look? (e) What do people buy in the two weeks after their first coffee subscription?
- 2 Write one association rule for a coffee chain. State its support and its confidence in words, then explain why a high confidence on its own does not justify launching a promotion.

ADVANCED

A rule reaches 90% confidence, and the item on the right-hand side appears in 92% of all receipts anyway. Is the rule interesting? Answer with a number and name the measure you used.

SUBMISSION

Add these answers to the same notebook as Practice 1. Together they form Task 3 of Lab 1.

1.4

Classification of Data Mining Systems

Four axes, and the disciplines that meet inside the field

1.4 Classification of Data Mining Systems

A multi-dimensional view

Data to be mined

Relational, warehouse and transactional data. Streams, time series and sequences. Text and web. Graphs, social and information networks. Spatial, spatiotemporal and multimedia data.

Knowledge to be mined

Characterisation, discrimination, association, classification, clustering, trend and outlier analysis. Descriptive and predictive. Several functions combined, and mining at several levels of abstraction.

Techniques utilised

Database and warehouse technology including OLAP. Machine learning, statistics, pattern recognition. Visualisation. High-performance and distributed computing.

Applications adapted

Retail, banking and insurance. Telecommunications. Fraud analysis and audit. Stock market analysis. Bio-data. Text and web mining.

Describing a data mining system means giving its position on all four axes. A tool that is excellent on one axis can be useless for your problem on another.

1.4 Classification of Data Mining Systems

A confluence of multiple disciplines

Machine learning

Models that improve with data

Statistics

Inference, uncertainty, significance

Pattern recognition

Features and similarity

Database technology

Storage, indexing, querying at scale

DATA MINING

where the eight meet

Algorithms

Complexity that stays usable

High-performance computing

Parallel and distributed execution

Visualisation

Making a pattern visible to a person

Applications

The domain question that starts it all

Each discipline supplies something the others lack. Statistics brings inference and the notion of significance, databases bring scale, and the applications bring the only thing that decides whether any of it mattered.

1.4 Classification of Data Mining Systems

Why one discipline was never going to be enough

Volume

Algorithms must remain usable on terabytes. An $O(n^2)$ method that is elegant on a thousand rows is unusable on a hundred million.

Dimensionality

A micro-array carries tens of thousands of attributes, and a text corpus more. Intuitions built in two or three dimensions stop working.

Complexity of the data

Streams, sequences, graphs, spatial and spatiotemporal data, text, web pages, heterogeneous and legacy systems, program code, simulation output.

New applications

Recommendation, real-time scoring, and invisible mining running inside products that never announce it.

TRADITIONAL DATA ANALYSIS ASSUMED

A small, clean, carefully sampled table, with a hypothesis written before the data was collected. Very little inside a firm looks like that.

1.5

Major Issues in Data Mining

What is still hard, and what is still contested

1.5 Major Issues in Data Mining

Issues (1): methodology and user interaction

Mining methodology

- Mining new and varied kinds of knowledge
- Mining knowledge in multidimensional space
- Handling noise, uncertainty and incompleteness, which is the normal state of business data
- Pattern evaluation, and constraint-guided mining
- Discovery in a networked environment
- Data mining as an interdisciplinary effort

User interaction

- Interactive mining, so that the analyst can steer the search instead of waiting for it
- Incorporation of background knowledge that the firm already holds
- Presentation and visualisation of results in a form a decision maker can interrogate
- Mining a business can audit, so that the reasoning is open to challenge and not only the output

A result that the decision maker cannot interrogate will not be used, however accurate it is. Interpretability is a business requirement before it is a research topic.

1.5 Major Issues in Data Mining

Issues (2): efficiency, diversity and society

Efficiency and scalability

- Efficiency and scalability of the algorithms themselves
- Parallel, distributed, stream and incremental methods
- Cost measured in money and in energy, not only in seconds

Diversity of data types

- Handling complex and structured types
- Mining dynamic, networked and global repositories
- Data that arrives faster than it can be stored

Data mining and society

- Social impact of mining decisions taken about people
- Privacy-preserving data mining
- Invisible data mining running inside ordinary products

Every dataset in this course describes people. In Vietnam, personal data is governed by the Personal Data Protection Law 2025 (Law 91/2025/QH15), in force from 1 January 2026, together with Decree 356/2025/ND-CP, which replaced Decree 13/2023/ND-CP. Ask what you are permitted to hold and to process before you ask what you are able to predict. Accuracy is never a defence for an unlawful or unfair use.

Quick check

1. Which task is data mining rather than retrieval?

- A. Counting yesterday's receipts for one store
- B. Finding item combinations nobody had thought to look for
- C. Listing all customers whose spend exceeds one million dong
- D. Printing a monthly revenue report from a cube

3. A rule has confidence 0.90 while its consequent appears in 92% of all receipts. What follows?

- A. The rule is strong and should be acted on
- B. The rule is barely better than the base rate, so lift is near or below one
- C. The support must also be 0.90
- D. Confidence cannot exceed the frequency of the consequent

2. In the KDD process, where does most of the project time usually go?

- A. Choosing the algorithm
- B. Running the algorithm
- C. Cleaning, integrating and transforming the data
- D. Presenting the results to management

4. Which pair correctly separates supervised from unsupervised work?

- A. Clustering is supervised, classification is unsupervised
- B. Both are supervised because both use data
- C. Classification uses labelled training records, clustering has no labels
- D. The difference is only the size of the dataset

Guided lab in Google Colab and submission

Task	What to produce
Task 1	Getting started: create a notebook named StudentID_FullName_DM_Lab01, add a text cell introducing yourself and stating in your own words what data mining is
Task 2	Load the Saigon Bean dataset, then report rows, columns, data types, missing values and the date range covered. Write two sentences on what the table can and cannot answer
Task 3	Practice 1 and Practice 2, one Markdown cell per exercise
Task 4	Place each of your five own questions about the dataset on the KDD pipeline, and name the functionality that would answer it
Task 5	Presentation and reproducibility: Markdown headings for every task, Runtime then Run all, correct sharing permission, link pasted into the LMS

SUBMISSION

Submit both ways before the next session: the file StudentID_FullName_DM_Lab01.ipynb on the LMS, and the notebook share link set so that anyone with the link can view. Late submissions lose 50% of the marks under the course regulations.

Summary

- 1 Data mining discovers interesting patterns and knowledge in data too large to read. Interesting means non-trivial, implicit, previously unknown and potentially useful.
- 2 It is a natural evolution of database technology, and it arrived when storage, integration and query power were already in place.
- 3 The KDD process runs from cleaning and integration through selection and transformation to mining, evaluation and presentation. Preparation dominates the timetable.
- 4 Mining applies to relational, warehouse and transactional data, and beyond that to streams, sequences, graphs, spatial data, text and the web.
- 5 The functionalities are characterisation and discrimination, association, classification and numeric prediction, clustering, outlier analysis, and sequence and network analysis.
- 6 The field is a confluence of machine learning, statistics, database technology, algorithms, high-performance computing, visualisation and the applications themselves.
- 7 The open issues are methodology, interaction, efficiency, the diversity of data, and the social questions that follow from mining data about people.

Reading and where the field publishes

For this lecture

- Han, Kamber & Pei, Data Mining: Concepts and Techniques, 3rd edition, Chapter 1. Library codes 160003005696 and 160003005699
- Wu, D. (2024), Data Mining with Python: Theory, Application, and Case Studies
- Đỗ Phúc, Giáo trình Khai phá dữ liệu, NXB ĐHQG-HCM, 2020

Going further

- Tan, Steinbach & Kumar, Introduction to Data Mining
- Hastie, Tibshirani & Friedman, The Elements of Statistical Learning
- Nguyễn Duy Nhất, Hồ Trung Thành, Lê Thị Kim Hiền, Khai phá dữ liệu trong kinh doanh, NXB ĐHQG-HCM, 2015

Conferences

- ACM SIGKDD, the flagship of the field since 1998
- IEEE ICDM, SIAM SDM, ECML-PKDD, PAKDD, WSDM
- Related venues: SIGMOD, VLDB, ICDE for data; ICML and NeurIPS for learning

Journals and finding papers

- Data Mining and Knowledge Discovery (DMKD), IEEE TKDE, ACM TKDD, SIGKDD Explorations
- Search through DBLP, Google Scholar and the UEL library databases



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UNIVERSITY OF ECONOMICS AND LAW

THANK YOU!

Next session: Chapter 2, Data Warehouse Technology

